



Department of Electrical and Electronic
Engineering
University of Peradeniya

EE404/572 Electric Power Systems

Power System Control and Modelling

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Module Contents

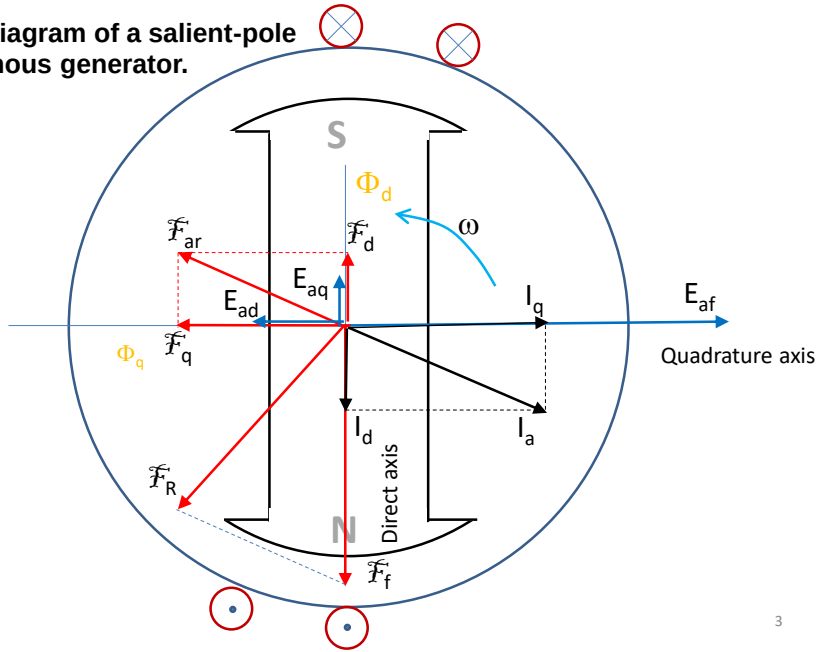
Power system control and modelling:

Modelling of power system equipment in dq frame, Models for frequency studies, Supplementary control actions, Hierarchical and decentralized control of power system

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d-q frame model of Salient-Pole Synchronous machine

Phasor diagram of a salient-pole synchronous generator.



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d-q frame model of Synchronous machine

$$\Phi_d = \frac{\mathcal{F}_d}{\mathcal{R}_d}$$

$$\Phi_q = \frac{\mathcal{F}_q}{\mathcal{R}_q}$$

$\mathcal{R}_d$ and $\mathcal{R}_q$ are reluctances
of direct and quadrature axis paths
 $\mathcal{R}_d < \mathcal{R}_q$

$$E_{ad} = -jX_{ad}I_d \quad E_{aq} = -jX_{aq}I_q$$

X_{ad} is the armature reaction reactance in the direct axis per phase

X_{aq} is the armature reaction reactance in the quadrature axis per phase

The total voltage induced in the stator = E'

$$E' = E_{af} + E_{ad} + E_{aq} = E_{af} - jX_{ad}I_d - jX_{aq}I_q$$

$$E' = V + R_a I_a + jX_l I_a = V + R_a I_a + jX_l (I_d + I_q)$$

$$E_{af} - jX_{ad}I_d - jX_{aq}I_q = V + R_a I_a + jX_l (I_d + I_q)$$

$$E_{af} = V + R_a I_a + j(X_l + X_{ad}) I_d + j(X_l + X_{aq}) I_q$$

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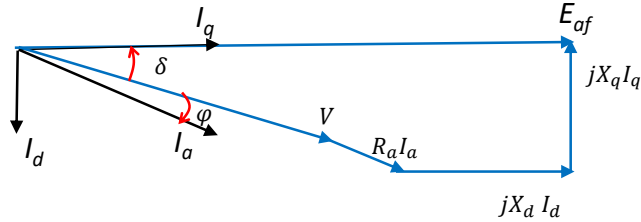
$$E_{af} = V + R_a I_a + jX_d I_d + jX_q I_q \text{ where } X_d \triangleq X_l + X_{ad} \text{ and } X_q \triangleq X_l + X_{aq}$$

X_d - Direct axis synchronous reactance

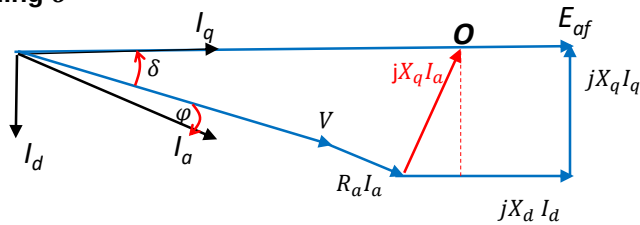
X_q - Quadrature axis synchronous reactance

Phasor diagram

Phasor diagram of a salient-pole synchronous generator.



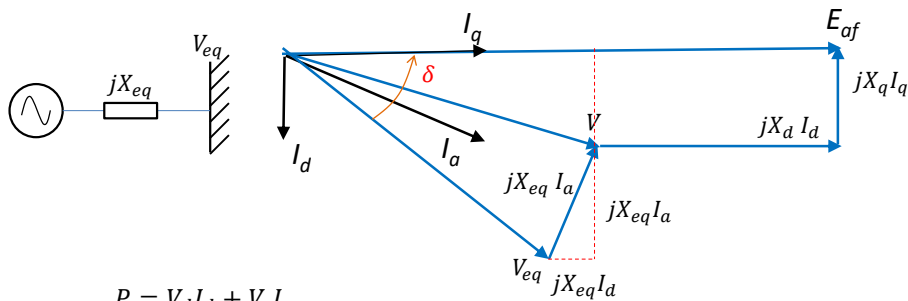
Determining δ



Determine point O

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POWER-ANGLE CHARACTERISTICS OF SALIENT-POLE MACHINES



$$P = V_d I_d + V_q I_q$$

$$I_d = \frac{E_{af} - V_{eq} \cos(\delta)}{X_{dT}}$$

$$I_q = \frac{V_{eq} \sin(\delta)}{X_{qT}}$$

$$X_{dT} = X_d + X_{eq}$$

$$X_{qT} = X_q + X_{eq}$$

$$P = \frac{E_{af} V_{eq}}{X_{dT}} \sin(\delta) + \frac{V_{eq}^2 (X_{dT} - X_{qT})}{2 X_{dT} X_{qT}} \sin(2\delta)$$

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Dq0 - Transformation

Basic Definitions

The dq0 transformation and its inverse are defined as

$$T_{\theta} = \frac{2}{3} \begin{bmatrix} \cos(\theta) & \cos(\theta - \frac{2\pi}{3}) & \cos(\theta + \frac{2\pi}{3}) \\ -\sin(\theta) & -\sin(\theta - \frac{2\pi}{3}) & -\sin(\theta + \frac{2\pi}{3}) \\ \frac{1}{2} & \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

$$T_{\theta}^{-1} = \begin{bmatrix} \cos(\theta) & -\sin(\theta) & 1 \\ \cos(\theta - \frac{2\pi}{3}) & -\sin(\theta - \frac{2\pi}{3}) & 1 \\ \cos(\theta + \frac{2\pi}{3}) & -\sin(\theta + \frac{2\pi}{3}) & 1 \end{bmatrix}$$

The angle θ is called here the reference angle or reference phase

The dq0 transformation maps three-phase signals in the **abc** reference frame to new quantities in a rotating **dq0** reference frame.

$$x_{abc} = [x_a, x_b, x_c]^T \quad x_{dq0} = [x_d, x_q, x_0]^T$$

$$x_{dq0} = T_{\theta} x_{abc}$$

$$x_{abc} = T_{\theta}^{-1} x_{dq0}$$

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Dq0 - Transformation

Basic Definitions

A fundamental property of the dq0 transformation is that it maps symmetric AC signals to constants.

For instance, consider a three-phase voltage source modeled as

$$v_a = A \cos(\omega_s t) + B,$$

$$v_b = A \cos\left(\omega_s t - \frac{2\pi}{3}\right) + B,$$

$$v_c = A \cos\left(\omega_s t + \frac{2\pi}{3}\right) + B.$$

with $\theta = \omega_s t$, it is easy to see that

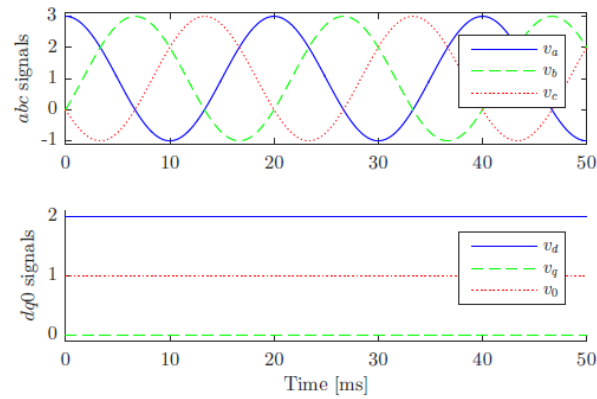
$$\begin{bmatrix} v_a \\ v_b \\ v_c \end{bmatrix} = \begin{bmatrix} \cos(\omega_s t) & -\sin(\omega_s t) & 1 \\ \cos(\omega_s t - \frac{2\pi}{3}) & -\sin(\omega_s t - \frac{2\pi}{3}) & 1 \\ \cos(\omega_s t + \frac{2\pi}{3}) & -\sin(\omega_s t + \frac{2\pi}{3}) & 1 \end{bmatrix} \begin{bmatrix} A \\ 0 \\ B \end{bmatrix}$$

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Dq0 - Transformation

Basic Definitions

Therefore $v_d = A$, $v_q = 0$, $v_0 = B$



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Dq0 - Transformation

Modeling Resistors

Assume a symmetrically configured three-phase resistor R , which is modeled as

$$\begin{bmatrix} v_a \\ v_b \\ v_c \end{bmatrix} = R \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} \quad \begin{array}{l} \text{Multiply both sides of} \\ \text{the equation by the} \\ \text{dq0 transformation } T_\theta \end{array} \quad \begin{bmatrix} v_d \\ v_q \\ v_0 \end{bmatrix} = R \begin{bmatrix} i_d \\ i_q \\ i_0 \end{bmatrix}$$

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Dq0 - Transformation

Modeling Inductors

symmetrically configured three-phase inductor L

$$\begin{bmatrix} v_a \\ v_b \\ v_c \end{bmatrix} = L \frac{d}{dt} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} \longrightarrow \begin{bmatrix} v_a \\ v_b \\ v_c \end{bmatrix} = L \frac{d}{dt} \left(T_\theta^{-1} \begin{bmatrix} i_d \\ i_q \\ i_0 \end{bmatrix} \right)$$

$$\begin{bmatrix} v_a \\ v_b \\ v_c \end{bmatrix} = L \left(\frac{d}{dt} T_\theta^{-1} \right) \begin{bmatrix} i_d \\ i_q \\ i_0 \end{bmatrix} + L T_\theta^{-1} \frac{d}{dt} \begin{bmatrix} i_d \\ i_q \\ i_0 \end{bmatrix}$$

$$\frac{d}{dt} T_\theta^{-1} = -T_\theta^{-1} \mathcal{W} \quad \mathcal{W} = \begin{bmatrix} 0 & \frac{d}{dt} \theta & 0 \\ -\frac{d}{dt} \theta & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

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Dq0 - Transformation

Modeling Inductors

$$\begin{bmatrix} v_a \\ v_b \\ v_c \end{bmatrix} = -L T_\theta^{-1} \mathcal{W} \begin{bmatrix} i_d \\ i_q \\ i_0 \end{bmatrix} + L T_\theta^{-1} \frac{d}{dt} \begin{bmatrix} i_d \\ i_q \\ i_0 \end{bmatrix}.$$

Multiply from the left by T_θ and rearrange terms to get

$$\frac{d}{dt} \begin{bmatrix} i_d \\ i_q \\ i_0 \end{bmatrix} = \mathcal{W} \begin{bmatrix} i_d \\ i_q \\ i_0 \end{bmatrix} + \frac{1}{L} \begin{bmatrix} v_d \\ v_q \\ v_0 \end{bmatrix}$$

The explicit expressions are

$$\begin{aligned} \frac{d}{dt} i_d &= \frac{d\theta}{dt} i_q + \frac{1}{L} v_d, \\ \frac{d}{dt} i_q &= -\frac{d\theta}{dt} i_d + \frac{1}{L} v_q, \\ \frac{d}{dt} i_0 &= \frac{1}{L} v_0. \end{aligned}$$

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Dq0 - Transformation

Modeling Capacitors

symmetrically configured three-phase Capacitor C

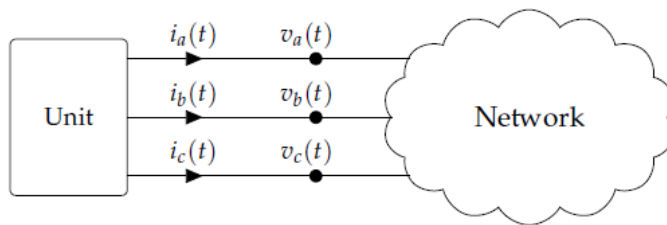
$$\frac{d}{dt} \begin{bmatrix} v_d \\ v_q \\ v_0 \end{bmatrix} = \mathcal{W} \begin{bmatrix} v_d \\ v_q \\ v_0 \end{bmatrix} + \frac{1}{C} \begin{bmatrix} i_d \\ i_q \\ i_0 \end{bmatrix}$$

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Dq0 - Transformation

Power in Terms of DQ0 Quantities

Assume a
general three-
phase unit



The instantaneous power
flowing from the unit into the
network at tir

$$P_{3\phi} = v_a i_a + v_b i_b + v_c i_c.$$

$$\begin{aligned} P_{3\phi} &= \begin{bmatrix} v_a & v_b & v_c \end{bmatrix} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} = \left(T_{\theta}^{-1} \begin{bmatrix} v_d \\ v_q \\ v_0 \end{bmatrix} \right)^T T_{\theta}^{-1} \begin{bmatrix} i_d \\ i_q \\ i_0 \end{bmatrix} \\ &= \begin{bmatrix} v_d & v_q & v_0 \end{bmatrix} \left(T_{\theta}^{-1} \right)^T T_{\theta}^{-1} \begin{bmatrix} i_d \\ i_q \\ i_0 \end{bmatrix}. \end{aligned}$$

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Dq0 - Transformation

Power in Terms of DQ0 Quantities

$$\left(T_{\theta}^{-1}\right)^T T_{\theta}^{-1} = \frac{3}{2} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

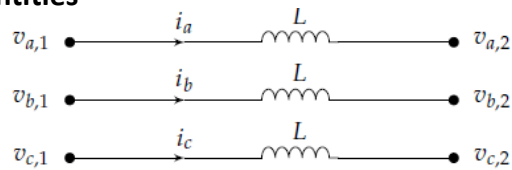
$$P_{3\phi} = \frac{3}{2} \begin{bmatrix} v_d & v_q & v_0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} i_d \\ i_q \\ i_0 \end{bmatrix} = \frac{3}{2} (v_d i_d + v_q i_q + 2v_0 i_0)$$

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Dq0 - Transformation

Energy in Terms of DQ0 Quantities

Assume a symmetrically configured three-phase inductor



The total energy stored in the inductor is $E = \frac{1}{2}L (i_a^2 + i_b^2 + i_c^2)$

$$\begin{aligned} E &= \frac{1}{2}L \begin{bmatrix} i_a & i_b & i_c \end{bmatrix} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix}^T = \frac{1}{2}L \left(T_{\theta}^{-1} \begin{bmatrix} i_d \\ i_q \\ i_0 \end{bmatrix} \right)^T T_{\theta}^{-1} \begin{bmatrix} i_d \\ i_q \\ i_0 \end{bmatrix} \\ &= \frac{1}{2}L \begin{bmatrix} i_d & i_q & i_0 \end{bmatrix} \left(T_{\theta}^{-1} \right)^T T_{\theta}^{-1} \begin{bmatrix} i_d \\ i_q \\ i_0 \end{bmatrix} \Rightarrow E = \frac{3}{4}L \begin{bmatrix} i_d & i_q & i_0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} i_d \\ i_q \\ i_0 \end{bmatrix} \\ &= \frac{3}{4}L (i_d^2 + i_q^2 + 2i_0^2) \end{aligned}$$

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Dq0 - Transformation

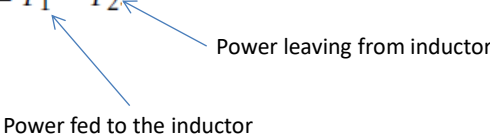
Energy in Terms of DQ0 Quantities

$$\frac{d}{dt}E = \frac{3}{2}L \left(i_d \frac{di_d}{dt} + i_q \frac{di_q}{dt} + 2i_0 \frac{di_0}{dt} \right)$$

$$\frac{d}{dt}E = \frac{3}{2}L \left[i_d \left(i_q \frac{d\theta}{dt} + \frac{1}{L}(v_{d,1} - v_{d,2}) \right) + i_q \left(-i_d \frac{d\theta}{dt} + \frac{1}{L}(v_{q,1} - v_{q,2}) \right) + 2i_0 \left(\frac{1}{L}(v_{0,1} - v_{0,2}) \right) \right]$$

$$\frac{d}{dt}E = \frac{3}{2} (v_{d,1}i_d + v_{q,1}i_q + 2v_{0,1}i_0) - \frac{3}{2} (v_{d,2}i_d + v_{q,2}i_q + 2v_{0,2}i_0) .$$

$$\frac{d}{dt}E = P_1 - P_2$$



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Dq0 - Transformation

Energy in Terms of DQ0 Quantities

Similarly, the energy stored in a symmetrically configured three-phase capacitor C is

$$E = \frac{3}{4}C (v_d^2 + v_q^2 + 2v_0^2)$$

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Dq0 - Transformation

Modeling General Linear Networks

Let x represent the state vector of the system in the abc reference frame, and use the compact notation

$$x_{abc} = [x_{a,1}, x_{b,1}, x_{c,1}, \dots, x_{a,m}, x_{b,m}, x_{c,m}]^T.$$

$$\frac{d}{dt}x_{abc} = Ax_{abc} + Bu,$$

$$\Lambda_\theta = \begin{bmatrix} T_\theta & & 0 \\ & \ddots & \\ 0 & & T_\theta \end{bmatrix}, \quad \Lambda_\theta^{-1} = \begin{bmatrix} T_\theta^{-1} & & 0 \\ & \ddots & \\ 0 & & T_\theta^{-1} \end{bmatrix}$$

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Dq0 - Transformation

Modeling General Linear Networks

$$\frac{d}{dt}(\Lambda_\theta^{-1})x_{dq0} + \Lambda_\theta^{-1}\frac{d}{dt}x_{dq0} = A\Lambda_\theta^{-1}x_{dq0} + Bu.$$

$$W_c = \begin{bmatrix} \mathcal{W} & & 0 \\ & \ddots & \\ 0 & & \mathcal{W} \end{bmatrix}, \quad \mathcal{W} = \begin{bmatrix} 0 & \frac{d}{dt}\theta & 0 \\ -\frac{d}{dt}\theta & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\frac{d}{dt}x_{dq0} = (\Lambda_\theta A \Lambda_\theta^{-1} + W_c)x_{dq0} + \Lambda_\theta Bu.$$

for the special case of symmetrically configured networks, it is typically true that

$$\Lambda_\theta A = A \Lambda_\theta,$$

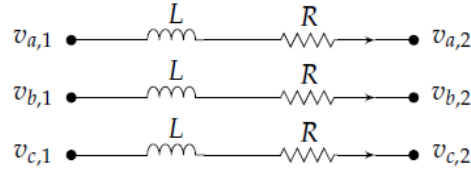
$$\frac{d}{dt}x_{dq0} = (A + W_c)x_{dq0} + \Lambda_\theta Bu$$

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Dq0 - Transformation

Modeling Linear Networks –An Example

Consider the circuit in Figure below



The network dynamics in the abc reference frame is

$$\frac{d}{dt} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} = -\frac{R}{L} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix} + \frac{1}{L} \begin{bmatrix} v_{a,1} - v_{a,2} \\ v_{b,1} - v_{b,2} \\ v_{c,1} - v_{c,2} \end{bmatrix}$$

$$\frac{d}{dt} i_{dq0} = (A + \mathcal{W}) i_{dq0} + \Lambda_\theta B u,$$

In this example $\Lambda_\theta = T_\theta$.

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Dq0 - Transformation

Modeling Linear Networks –An Example

$$\frac{d}{dt} \begin{bmatrix} i_d \\ i_q \\ i_0 \end{bmatrix} = (A + \mathcal{W}) \begin{bmatrix} i_d \\ i_q \\ i_0 \end{bmatrix} + T_\theta B \begin{bmatrix} v_{a,1} - v_{a,2} \\ v_{b,1} - v_{b,2} \\ v_{c,1} - v_{c,2} \end{bmatrix}$$

$$\frac{d}{dt} \begin{bmatrix} i_d \\ i_q \\ i_0 \end{bmatrix} = \left(-\frac{R}{L} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & \frac{d}{dt}\theta & 0 \\ -\frac{d}{dt}\theta & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \right) \begin{bmatrix} i_d \\ i_q \\ i_0 \end{bmatrix} + \frac{1}{L} T_\theta \begin{bmatrix} v_{a,1} - v_{a,2} \\ v_{b,1} - v_{b,2} \\ v_{c,1} - v_{c,2} \end{bmatrix}$$

$$\frac{d}{dt} \begin{bmatrix} i_d \\ i_q \\ i_0 \end{bmatrix} = \begin{bmatrix} -\frac{R}{L} & \frac{d}{dt}\theta & 0 \\ -\frac{d}{dt}\theta & -\frac{R}{L} & 0 \\ 0 & 0 & -\frac{R}{L} \end{bmatrix} \begin{bmatrix} i_d \\ i_q \\ i_0 \end{bmatrix} + \frac{1}{L} \begin{bmatrix} v_{d,1} - v_{d,2} \\ v_{q,1} - v_{q,2} \\ v_{0,1} - v_{0,2} \end{bmatrix}$$

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Dq0 - Transformation

Modeling Linear Networks –An Example

Choosing the reference frame such that $\frac{d}{dt}\theta = \omega_s$

$$\frac{d}{dt} \begin{bmatrix} i_d \\ i_q \\ i_0 \end{bmatrix} = \begin{bmatrix} -\frac{R}{L} & \omega_s & 0 \\ -\omega_s & -\frac{R}{L} & 0 \\ 0 & 0 & -\frac{R}{L} \end{bmatrix} \begin{bmatrix} i_d \\ i_q \\ i_0 \end{bmatrix} + \frac{1}{L} \begin{bmatrix} v_{d,1} - v_{d,2} \\ v_{q,1} - v_{q,2} \\ v_{0,1} - v_{0,2} \end{bmatrix}$$

The resulting model is linear and time-invariant

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Dq0 - Transformation

Modeling Linear Networks –An Example

Choosing the reference frame such that $\frac{d}{dt}\theta = \omega_s$

$$\frac{d}{dt} \begin{bmatrix} i_d \\ i_q \\ i_0 \end{bmatrix} = \begin{bmatrix} -\frac{R}{L} & \omega_s & 0 \\ -\omega_s & -\frac{R}{L} & 0 \\ 0 & 0 & -\frac{R}{L} \end{bmatrix} \begin{bmatrix} i_d \\ i_q \\ i_0 \end{bmatrix} + \frac{1}{L} \begin{bmatrix} v_{d,1} - v_{d,2} \\ v_{q,1} - v_{q,2} \\ v_{0,1} - v_{0,2} \end{bmatrix}$$

The resulting model is linear and time-invariant

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Dq0 - Transformation

Comparison of Time-Varying Phasors and DQ0 Models

Assume a unit with
balanced three-phase
AC voltages given by

$$\begin{aligned}v_a(t) &= A(t) \cos(\omega_s t + \psi(t)), \\v_b(t) &= A(t) \cos\left(\omega_s t + \psi(t) - \frac{2\pi}{3}\right) \\v_c(t) &= A(t) \cos\left(\omega_s t + \psi(t) + \frac{2\pi}{3}\right)\end{aligned}$$

These voltages may be represented
by the time-varying phasor

$$V(t) = \frac{A(t)}{\sqrt{2}} e^{j\psi(t)}$$

$$\begin{aligned}\begin{bmatrix} v_a \\ v_b \\ v_c \end{bmatrix} &= \begin{bmatrix} A(t) \cos(\omega_s t + \psi(t)) \\ A(t) \cos(\omega_s t + \psi(t) - \frac{2\pi}{3}) \\ A(t) \cos(\omega_s t + \psi(t) + \frac{2\pi}{3}) \end{bmatrix} \\ &= \begin{bmatrix} \cos(\omega_s t) & -\sin(\omega_s t) & 1 \\ \cos(\omega_s t - \frac{2\pi}{3}) & -\sin(\omega_s t - \frac{2\pi}{3}) & 1 \\ \cos(\omega_s t + \frac{2\pi}{3}) & -\sin(\omega_s t + \frac{2\pi}{3}) & 1 \end{bmatrix} \begin{bmatrix} A(t) \cos(\psi(t)) \\ A(t) \sin(\psi(t)) \\ 0 \end{bmatrix} = T_{\omega_s t}^{-1} \begin{bmatrix} v_d \\ v_q \\ v_0 \end{bmatrix}\end{aligned}$$

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Dq0 - Transformation

Comparison of Time-Varying Phasors and DQ0 Models

$$\begin{aligned}v_d(t) &= A(t) \cos(\psi(t)) \\v_q(t) &= A(t) \sin(\psi(t)), \\v_0(t) &= 0.\end{aligned}$$

As an example, consider again the
symmetrically configured three-phase
inductor, which dynamic model

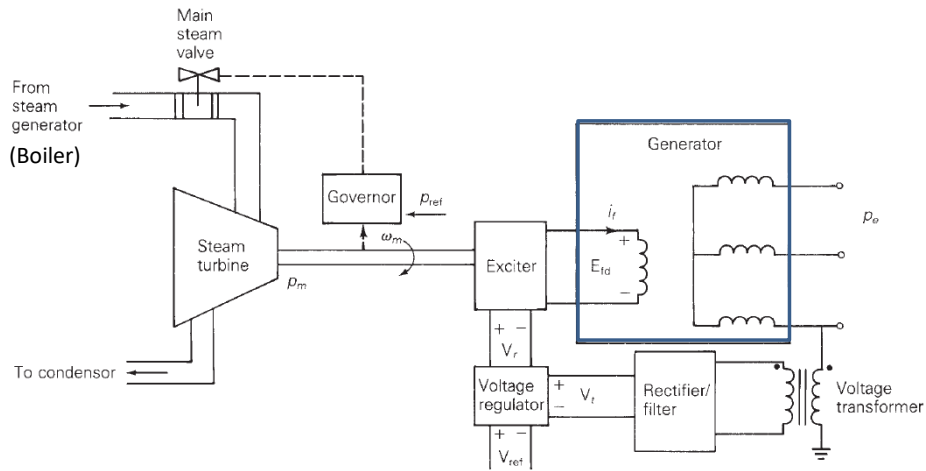
$$\begin{aligned}v_d &= -\omega_s L i_q + L \frac{d}{dt} i_d, \\v_q &= \omega_s L i_d + L \frac{d}{dt} i_q.\end{aligned}$$

The equivalent expressions for the
quasi-static model are

$$\begin{aligned}v_d &= -\omega_s L i_q, \\v_q &= \omega_s L i_d.\end{aligned}$$

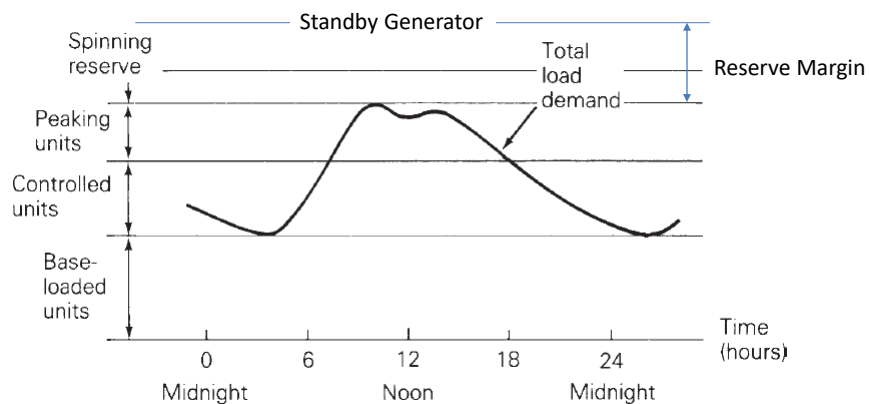
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Power System Control



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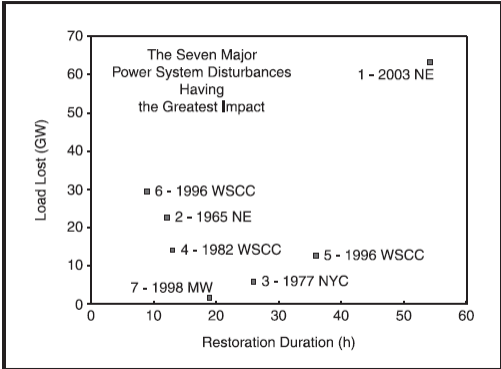
Typical Load Cycle



The central control center monitors information including area frequency, generating unit outputs, and tie-line power flows to interconnected areas. This information is used by *automatic load-frequency control (LFC)* in order to maintain area frequency at its scheduled value (50 or 60 Hz) and net tie-line power flow out of the area at its scheduled value.

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Major Disturbances



Recent Blackouts in Sri Lanka

Date	Time taken to restore
9-Oct-09	7 Hr 57 min
27-Sep-15	4 Hr 5 min
25-Feb-16	2 Hr 28 min
3-Mar-16	7 Hr 16 min

Overcoming Restoration Challenges Associated with Major Power System Disturbances

Reactive power unbalance, involving generator under excitations, sustained overvoltage, and switched capacitors/reactors.

load and generation unbalance, including responses to sudden increases in load, and underfrequency load shedding.

inadequate load and generation coordination, including lack of black-start capability, problems with switching operations, line overloads, and control center coordination.

monitoring and control inadequacies, including communication, supervisory control and data acquisition (SCADA) system capabilities, computer overloading, display capabilities, simulation tools, and system status determination.

protective systems, including interlocking schemes, synchronization and synchrocheck, standing phase angles, and problems with other types of relays

depletion of energy storage, including low-pressure compressed air/gas and discharged batteries.

system restoration plan inadequacies, including lack of planned procedure, outdated procedure, procedure not being followed, inadequate training, and, incredibly, lack of standard communication vocabulary.

Typical Load Cycle

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FREQUENCY CONTROL

- Electricity has to be generated the instant it is used.
- Automatic control systems are necessary in all larger power systems.

SYNCHRONOUS GRIDS

- A power system connected by AC lines and transformers constitutes a **synchronous grid**.
- There can only be one frequency in a synchronous grid.

FREQUENCY CONTROL

FREQUENCY RANGE

- The frequency of a large power system **must** be kept around the nominal value!
- Certain frequencies can harm important equipment in the power system.
 - Harmonic vibrations in turbine blades and shafts.
 - Heating of generators and transformers.
- Some loads might be disturbed.
- Automatic control systems:
 - Normal operation (e.g. 49.9 - 50.1 Hz)
 - Disturbance reserve (e.g. 49.5 - 49.9 Hz)
- Disconnection of interruptible loads (export on HVDC lines, electric boilers, heat pumps, pumped storage hydro (e.g.: 49.0 - 49.8 Hz)
- Automatic load shedding (e.g.: 48.0 - 48.8 Hz)
- Manual load shedding (rotating load curtailment (e.g.: < 48.0 Hz)

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FREQUENCY CONTROL

FREQUENCY CONTROL SYSTEMS

- Primary control
 - Response time: seconds
 - Automatic control system
- Secondary control
 - Response time: minutes
 - Automatic control system
- Tertiary control
 - Response time: 10–15 minutes
 - Manually activated

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FREQUENCY CONTROL

PRIMARY CONTROL

The objective of the primary control is to maintain the balance between generation and load.

In control theory, the primary control corresponds to a proportional controller (P controller).

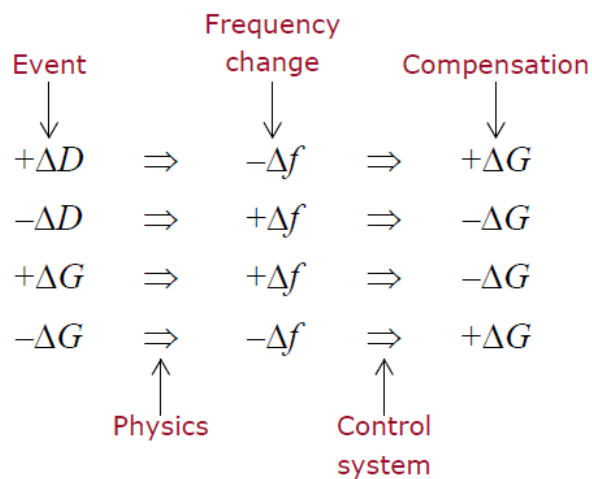
Example: Load Increases

- System in balance initially.
- Load increases, but generation is unaffected.
- Energy is taken from the kinetic energy of all synchronous machines.
- The synchronous machines will rotate slower.
- The frequency decreases.
- Frequency sensors in some power plants detect the frequency decrease and increase generation in that power plant until the frequency is stable again.

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FREQUENCY CONTROL

PRIMARY CONTROL



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FREQUENCY CONTROL

PRIMARY CONTROL

Definition 1: The gain states the change in generation for specific change in frequency:

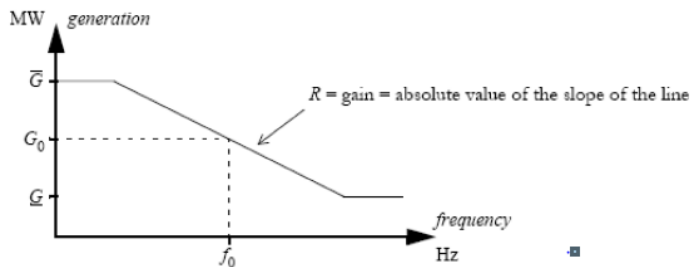
$$G = G_0 - R(f - f_0),$$

where

G_0 = base generation,

R = gain,

f_0 = nominal frequency.



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FREQUENCY CONTROL

PRIMARY CONTROL

GAIN

In general, it is better to just consider the changes, i.e., $\Delta G = R \cdot \Delta f$, and then think!

Example 1:

- Generation 1 000 MW
- Load 1 000 MW
- Gain 200 MW/Hz
- Frequency 49.98 Hz.

What is the new frequency going to be if the load increases by 10 MW?

$$\Delta f = \frac{\Delta G}{R} = \frac{10}{200} = 0.05 \text{ Hz},$$

Solution

- Load increase $\rightarrow$ Frequency decrease
- Hence, the new frequency is
 $f = 49.98 - 0.05 = 49.93 \text{ Hz}.$

FREQUENCY CONTROL

LIMITATIONS IN PRIMARY CONTROL

When designing a power system, it is important to consider that there are limitations to the primary control:

- The power plants participating in primary control have limited primary control reserves.
- The gain must be fairly evenly distributed in the system, because otherwise the transmission grid might be overloaded.

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FREQUENCY CONTROL

LIMITATIONS IN PRIMARY CONTROL

EXAMPLE

Power system in balance, nominal frequency (50 Hz). Two power plants participating in primary control:

- **Unit 1:** 500 MW capacity, current generation 420 MW, gain 300 MW/Hz
 - **Unit 2:** 100 MW capacity, current generation 75 MW, gain 300 MW/Hz
- 85 MW of generation is lost in another unit.

Solution

Same gain → both units increase their generation by the same amount, i.e., $85/2 = 42.5$ MW.

Unit 2 can only increase generation by 25 MW! When the frequency is about 49.916 Hz unit 2 will stop regulating.

- To restore balance, unit 1 must increase its generation by 60 MW.

- $\Delta f = \frac{\Delta G}{R} = \frac{60}{300} \text{ Hz} = 0.2 \text{ Hz}.$

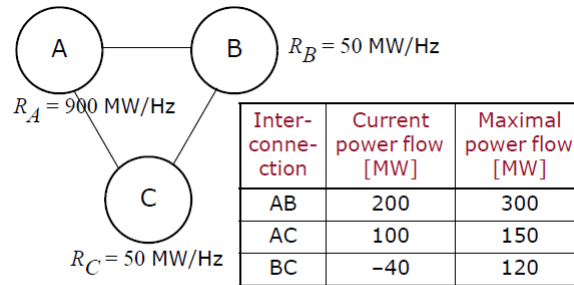
- New frequency $f = 50 - 0.2 = 49.8 \text{ Hz}.$

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FREQUENCY CONTROL

LIMITATIONS IN PRIMARY CONTROL

EXAMPLE



- 240 MW generation is lost in area B.

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FREQUENCY CONTROL

LIMITATIONS IN PRIMARY CONTROL

Solution

- Generation has to increase by 240 MW.
- The increase is distributed between the areas according to their share of the total gain, $R_{ABC} = 1\,000 \text{ MW/Hz}$.

$$\Delta G_A = \frac{R_A}{R_{ABC}} \cdot 240 = 216 \text{ MW.}$$

$$\Delta G_B = \frac{R_B}{R_{ABC}} \cdot 240 = 12 \text{ MW.}$$

$$\Delta G_C = \frac{R_C}{R_{ABC}} \cdot 240 = 12 \text{ MW.}$$

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FREQUENCY CONTROL

LIMITATIONS IN PRIMARY CONTROL

Solution Area A

- Same load, increased generation $\Rightarrow$ increased export.
- However, only 150 MW export capacity available $\Rightarrow$ both transmission lines will eventually be overloaded.
- The area is isolated from the rest of the system $\Rightarrow$ reduced export 300 MW (compared to before the fault).
Reduced export equivalent to a load reduction!
- Frequency increase $\Delta f_A = 300/900 = 0.33$ Hz.

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FREQUENCY CONTROL

LIMITATIONS IN PRIMARY CONTROL

Solution

Areas B and C

- Total deficit 540 MW (240 MW lost generation, 300 MW lost import).
- Equal gain $\Rightarrow$ both areas increase generation by 270 MW.

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FREQUENCY CONTROL

LIMITATIONS IN PRIMARY CONTROL

Solution

Area C

- Decreased import 100 MW, increased generation 270 MW $\Rightarrow$ increased export by 170 MW.
- Only 80 MW export capacity $\Rightarrow$ transmission line overloaded.
- The area is isolated from the rest of the system $\Rightarrow$ deficit 60 MW compared to before the fault (lost 100 MW import and 40 MW export).
- Frequency decrease $\Delta f_C = 60/50 = 1.2$ Hz.

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FREQUENCY CONTROL

LIMITATIONS IN PRIMARY CONTROL

Solution Area B

- The area is isolated from the rest of the system $\Rightarrow$ deficit 480 MW compared to before the fault (lost 240 MW import and 240 MW generation).
- Frequency decrease $\Delta f_B = 480/50 = 9.6$ Hz.

Summary

- All transmission lines overloaded
- New frequency in area A: 50.33 Hz
- New frequency in area B: 40.4 Hz (!)
- New frequency in area C: 48.8 Hz

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FREQUENCY CONTROL

PRIMARY CONTROL IN RESTRUCTURED ELECTRICITY MARKETS

The system operator is responsible for primary control, but does not have sufficient generation capacity. Two solutions:

- Require producers to supply certain amounts of gain to be allowed to connect to the grid.
- Pay producers to supply gain.

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FREQUENCY CONTROL

PRIMARY CONTROL PRICING

The players who supply primary control to the system can be reimbursed for

- provision of primary control capacity (\$ per MW/Hz)
- utilisation of primary control (\$/MWh)
The energy price may for example be obtained from the real-time market.

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FREQUENCY CONTROL

COSTS OF PRIMARY CONTROL

The costs of supplying primary control capacity depends on the generation technology.

- **Wind power, photovoltaics:** Non-dispatchable units $\Rightarrow$ very high costs.
- **Hydro power:** Negligible fuel costs, small efficiency losses $\Rightarrow$ low costs.
- **Large thermal units:** Fuel costs, efficiency losses $\Rightarrow$ high costs.
- **Nuclear power plants:** Designed to be operated at installed capacity $\Rightarrow$ quite high costs.
- **Small thermal units:** Fuel costs, small efficiency losses $\Rightarrow$ rather high costs.

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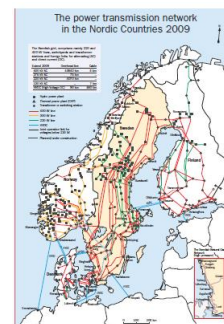
FREQUENCY CONTROL

PRIMARY CONTROL RESERVES

The amount of capacity available for the primary control depends on the requirements for the system.

Example: The Nordic system

- Total gain $R = 6\,000 \text{ MW/Hz}$
- Normal operation:
frequency within $50 \pm 0.1 \text{ Hz}$
- The necessary normal operation reserves are
 $\Delta G = R \cdot \Delta f = 600 \text{ MW}.$



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FREQUENCY CONTROL

SECONDARY CONTROL

The objective of the secondary control is to

- restore the frequency to its nominal value,
- release used primary control reserves,
- reduce the integrated time error.

In control theory, the secondary control corresponds to an integral controller (I controller).

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FREQUENCY CONTROL

SECONDARY CONTROL

Secondary control operates by changing the base generation* of the system.

- Change generation in a power plant that is not participating in the primary control.
- Change base generation in a power plant that is participating in the primary control.

* By which we mean the generation at nominal frequency.

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FREQUENCY CONTROL

SECONDARY CONTROL

- If the frequency is **less** than the nominal then some generation capacity needs to be started, or the load decreased.
- If the frequency is **larger** than the nominal then some generation capacity needs to be stopped, or the load increased.



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FREQUENCY CONTROL

CONTROLLING THE FREQUENCY

Secondary control can be performed by

- Automatic control systems (**AGC** - Automatic Generation Control).
- Manual control (using real-time trading).

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FREQUENCY CONTROL

TERTIARY CONTROL

TERTIARY CONTROL is any automatic or manual change in the working points of generators or loads participating, in order to:

- guarantee the provision of an adequate SECONDARY CONTROL RESERVE at the right time,
- distribute the SECONDARY CONTROL POWER to the various generators in the best possible way, in terms of economic considerations.

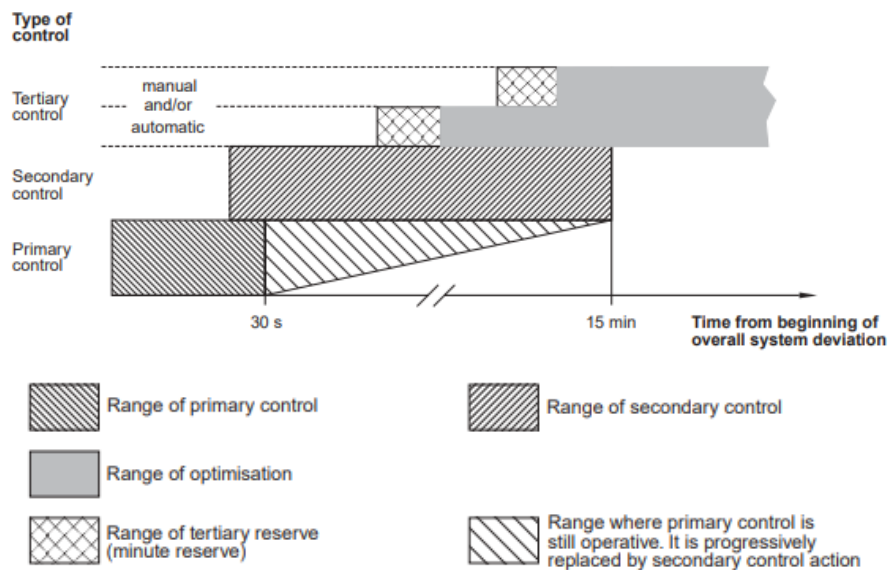
Changes may be achieved by:

- connection and tripping of power (gas turbines, reservoir and pumped storage power stations, increasing or reducing the output of generators in service);
- redistributing the output from generators participating in secondary control;
- changing the power interchange program between interconnected undertakings;
- load control (e.g. centralized telecontrol or controlled LOAD-SCHEDDING).

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FREQUENCY CONTROL

OVERALL PICTURE



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FREQUENCY CONTROL

TIME ERROR

- The integrated time error is the deviation between normal time and synchronous time (i.e., a clock controlled by the grid frequency).
- The time deviation during a certain period in a 50 Hz system is given by

$$t_i = \int_0^t \frac{(f(\tau) - 50)}{50} d\tau.$$

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FREQUENCY CONTROL

CONTROLLING THE TIME ERROR

- To reduce a **negative** time error, the frequency must be larger than the nominal value for some time.
- To reduce a **positive** time error, the frequency must be less than the nominal value for some time (generally not a problem).
- The time error is not considered very important for the operation of the power system.

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FREQUENCY CONTROL

TIME ERROR - Example

- Current time deviation: -2 s.
- Frequency the next five minutes: 50.1 Hz.

What is the time deviation after this period?

Solution:

- During each second the grid clock will pass 0.1 cycles more than normal time.
- Grid clock will gain $0.1/50 \cdot 300 = 0.6$ s compared to normal time.
- Time error after five minutes
 $t_i = -2 + 0.6 = -1.4$ s.

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FREQUENCY CONTROL

TIME ERROR - Practical usage

The time error can be used to calculate the real-time trading of a balance responsible player participating in primary control:

Example:

- Power Ltd. is a balance responsible player.
- During the trading period 9-10 am. they have sold $3\,000$ MWh to the power pool, and $3\,560$ MWh directly to consumers.

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FREQUENCY CONTROL

TIME ERROR - Practical usage

- The generation in the base power plants of the company (i.e., power plants not participating in primary control) was 4 650 MWh.
- Power Ltd. also has power plants participating in the primary control
 - Base generation (G_0) 2 000 MW
 - Gain (R) 1 000 MW/Hz
 - Neglect limitations in primary control reserves
- The time error 9:00 is -0.42 and the time error 10:00 is $+0.30$.

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FREQUENCY CONTROL

TIME ERROR - Practical usage

- The relation between frequency and time deviation is linear $\Rightarrow$ Assume constant frequency during the hour.
- The grid clock changes by $(f - 50)/50$ s every second.
- The total change during 3 600 s is $+0.72$ s $\Rightarrow f = 50.01$ Hz.
- At $f = 50.01$ Hz, the power plants participating in primary control generate $G = G_0 - R(f - f_0) = 1\,990$ MW.

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FREQUENCY CONTROL

TIME ERROR - Practical usage

- Hence, Power Ltd. is buying 10 MWh from the system operator during the real-time trading.
- The imbalance of Power Ltd. is
 $4\,650 + 1\,990 - 10 - 3\,000 - 3\,560 = 70$ MWh,
i.e., the company has sold 70 MWh
imbalance power to the system operator.